



Indian Institute of Technology Bhilai
Department of Computer Science and Engineering
Database Management Systems
(CSL303/MAL505/CS254)

Mid-Semester, Date: September 24, 2025

Timing: 09:30 AM to 11:30 AM

2025-26-M Semester

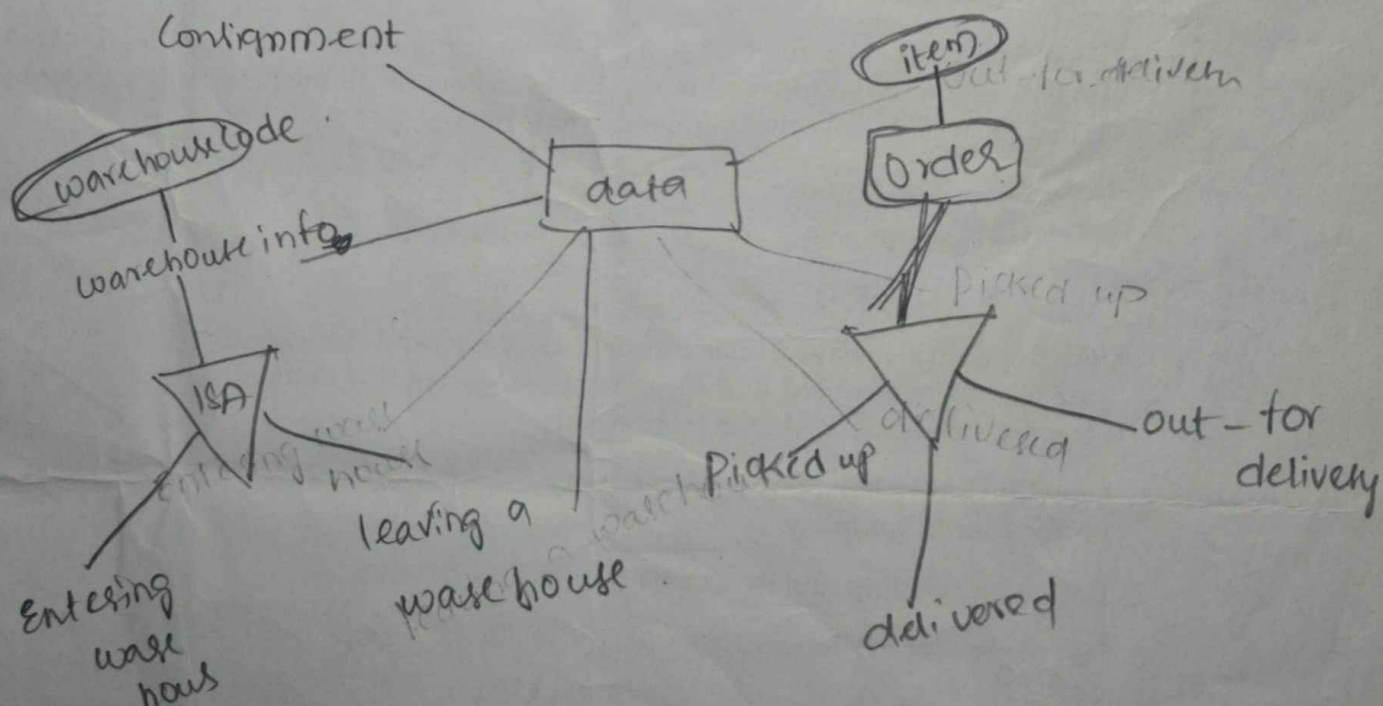
Max mark: 50

Consider the scenario of a courier company trying to build a record keeping system for its operation. The following are the requirements stated about the operation of the company

- Each consignment has an alphanumeric unique consignment id, sender's name, sender's address, sender's phone, receiver's name, receiver's address, receiver's phone and cost paid by the sender.
 - The courier company has few warehouses throughout the country. The warehouses have a unique 5 character alphabet code along with their name, city and pin.
 - The event of the item entering a warehouse and leaving a warehouse has to be recorded along with warehouse info, date-time of the event and a remark.
 - Each item goes through a list of warehouses before it reaches the last warehouse at the receiver's location. The list of warehouses that an item goes through does not contain any repetitions.
 - A item can have a special event called "Out for Delivery" when it exits the nearest warehouse of receiver's location. This event should be recorded similar to other events with date-time and remark.
 - Each warehouse is connected to a few other warehouses. For example, if warehouse1 is connected to warehouse2 and warehouse3, all the items leaving warehouse1 can have only the following events next
 - Enter Warehouse2
 - Enter Warehouse3
 - Out for delivery
 - An item can also have events called "Picked Up" and "Delivered". These events are also to be recorded in the same manner except that there are no warehouse information associated with these events.
1. Draw the Entity Relationship diagram for the database that would store the data for the courier company. (5)
 2. Derive the relations and its schema from the ER Diagram. (4)
 3. List the functional dependencies for the relational schema that needs to be satisfied. (5)

$\alpha \rightarrow \beta$ $\beta \subset \alpha$ α is Superkey

4. Perform Normalization and show the list of relational schema in 3NF and in BCNF. Specify the foreign key relationships as well. Write your steps explicitly. (6)
5. Write the necessary DDL query in SQL for creating the relational schema with the obtained constraints. (4)
6. Write the collections that you will create if you need to store the data of the courier company in a NoSQL database. Give the document structure for each of the collection with example. (6)
7. Write SQL query AND MongoDB query to get the following results from the relational schema or Collections (respectively) produced.
 - (a) Given a consignment ID show the tracking details of the item. Show the city of the warehouses it is/was in, upto the current status along with date-time and remark. (4)
 - (b) Find the list of warehouses in decreasing order of the number of items processed by them. (4)
 - (c) Find the total number of items processed by the courier company as a whole with two given dates. The items still undelivered must also be counted. (4)
 - (d) Find the number of items that are currently in possession of the courier company (i.e. not yet delivered) (4)
 - (e) Find out the number of warehouses each city has. Do not consider the cities which does not have any warehouses. (4)
8. Find all items which exits some warehouse1 and enters another warehouse2 next, where warehouse1 and warehouse2 are not connected. (5 (bonus))





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2025-26-M Semester

Max mark: 60

1. A university database has the following relations:

Student(Roll, Name, Dept), Course(CID, Title, Dept), Registered(Roll, CID, Semester)

Write relational algebra expressions to find:

- (a) Names of students registered in all courses offered by their own department. (3)
(b) Students who are registered for more than three courses in a semester. (3)

2. A school maintains the relation:

ClassInfo(ClassID, TeacherName, Subject, RoomNo, TeacherPhone)

Given:

TeacherPhone $\rightarrow$ TeacherName, TeacherName $\rightarrow$ Subject, ClassID $\rightarrow$ RoomNo, TeacherName

Normalize the relation to BCNF and show each decomposition step.

3. Given the schema of a ride-sharing service:

Driver(DID, Name, Rating)

Ride(RideID, DID, Source, Destination, Fare, Date)

Write SQL queries to:

- (a) Display drivers who have completed more than 10 rides in the last month. (3)
(b) Display drivers who have the highest average fare across all their rides. (2)

4. In a movie-booking platform:

Movie(MID, Title, Genre, ReleaseYear)

Show(SID, MID, Theatre, ShowTime)

Booking(BID, SID, UserID, SeatsBooked)

Write an SQL query to list the top 5 movies with the highest number of total seats booked, ordered by popularity.

5. A logistics company wants to track vehicles, drivers, shipments, and delivery times.

- Each shipment is transported by one vehicle and one driver.

→ SID VID DID

driver (DID, VID)

- A driver can drive multiple vehicles over time.
- A shipment may pass through multiple hubs with timestamps.

Tasks:

- Draw an ER diagram capturing these constraints. (4)
- Convert the ER diagram to relational schema. (3)
- Identify anomalies and normalize the schema to 3NF. (3)

6. A food-delivery company stores orders in:

Order(OID, CustomerID, RestaurantID, OrderTime, TotalAmount)

OrderItem(OID, ItemID, Quantity, Price)

- Write a relational algebra query to find restaurants with average order value greater than Rs. 500. (3)
- Write an SQL query to list the top 3 restaurants with the highest revenue in the last 7 days. (4)
- Suggest indexing strategies to optimize both queries. (3)

7. What are the advantages of Reader-Writer locks for latches compared to simple blocking OS mutexes. (4)

8. Consider an implementation of 7-way external merge sort.

- How many buffer pool pages are required for a basic implementation. (2)
- Consider that the I/O cost of 1 page is k times the amount of time required in computation (merge) of the tuples in page. What would be the theoretically optimum number of buffer pool pages required? (3)
- If you decide to implement double buffering with the basic implementation of 7-way external merge sort, what would be the number of buffer pages required? (2)
- If you have B buffer pages available, derive an equation for finding out the optimum value for m for doing m -way merge sort with double buffering. (2)

9. Consider executing a join operation on two tables R and S . R has 800 pages and 75000 tuples whereas S has 450 pages and 42000 tuples. The cost for index lookup on join attribute is 20 unit for R and 18 units for S on an average.

- Find the minimum cost of I/O required, when Naive Nested loops are used to perform join. (3)
- Find the minimum cost of I/O required, when Block Nested loops are used to perform join. (3)
- Find the minimum cost of I/O required, when Index Nested loops are used to perform join. (3)